PROJECT PROPOSAL

(REVISED VERSION)

GROUP 3 MEMBERS:
1.) UDTOHAN 2.) CONCEPCION 3.) MATIRA

1.) Problem Statement

- Our system is designed forbudget-conscious students and shoppers who need to stick to a strict spending limit before paying at the register. Relying on mental math or standard phone calculators is really inefficient because it's easy to forget what you've already added, and one wrong press ruins your whole total with no way to double-check. A common situation is when students buy materials for group projects. They throw items into the basket while guessing the running cost, only to get embarrassed at the cashier when they run out of money and have to awkwardly put items back, holding up the line. Almost every student has dealt with this kind of "cart shock." Our program fixes this problem by letting users track their items clearlyand catch budget overspends before they even step up to the register.

2.) Project Objectives

- Ourgoal is tobuild a user-friendly Python console program that automatically tracks total costs and warns you if you're going over your limit. By using the built-in Python tools/functions, we are creating a Smart Pre-Checkout Tool. Standard calculators are frustrating because they only show raw numbers without item names, making it hard to spot typos. Meanwhile, actual cashier terminals only tally items when it's already too late to change your mind. Our Python system bridges thisgap by giving shoppers full control and an organized breakdown beforehand.

3.) Planned Features

- **Preset Item Menu:** A built-in catalog of 5 common school supplies using Python dictionary lookup, allowing users to select items quickly using number codes.
- **Custom Item Entry with Validation:** An option to manually type in unlisted items and prices, backed by input validation guardrails to block negative numbers, letters, or invalid inputs from crashing the program.
- **Quantity Multiplier:** Allows users to input the number of pieces they want for any selected item, automatically multiplying the cost to calculate the subtotal.

- **Live Budget Tracker & Warning System:** Real-time budget monitoring that calculates running totals against an initial budget limit and alerts the user immediately if an item exceeds their remaining balance.
- **Receipt Text File Export:** Automatically generates and saves a downloadable receipt.txt file containing the complete itemized purchase summary at the end of the session.
- **Modular Code Design:** Built cleanly using Python functions (def) to organize the program into structured, professional logic blocks.

4.) Planned Inputs and Outputs

- **Planned Inputs:**
- **Initial Budget Limit:** The maximum spending amount set by the user at the start of the program.
- **Menu Choices:** Number codes entered by the user to choose preset items, add custom items, or exit the session.
- **Custom Item Data:** Custom item names and numeric prices entered manually when selecting unlisted items.
- **Item Quantities:** The number of pieces or units specified for each chosen item.

- **Planned Outputs:**
- **Interactive MenuDisplay:** A clean, numbered list of available preset and custom options shown on the console.
- **Live Budget Updates:** Real-time feedback showing the updated running total and remaining budget after each entry.
- **Budget Warning Alerts:** Instant notifications if an added item exceeds the user's set spending limit.
- **Final Itemized Summary:** A receipt-style summary displayed on the console and automatically exported as a text file (receipt.txt), listing all items, quantities, sub-totals, and the final total cost.

Flowchart Logic



